13

Large worlds of mathematical structures

A further exploration of large categories, based on sets and functions.

We will now develop some categories of sets-with-structure, starting with monoids. The morphisms will then use the idea of "structure-preserving maps".

13.1 Monoids

As we said at the beginning of Chapter 12, we previously zoomed in and looked at an *individual* monoid as a (small) one-object category, but we can also zoom out and look at the *totality* of monoids. In this large category, the objects are monoids themselves, and the morphisms are functions between them.

Our aim now is to come up with a definition of "function between monoids" that is a sensible form of relationship for monoids. Recall[†] that a monoid has

- data: a set of objects
- structure: an identity and a binary operation
- properties: unitality and associativity.

So it can be broadly summed up as a "set equipped with extra structure (satisfying some properties)". Whenever we have "sets-with-structure" we can look for "sensible" functions between them, that is, functions that respect that structure. An arbitrary function might not respect the structure, in which case it wouldn't be such a useful way of relating two different examples to each other.

For example, let's start with the monoid of natural numbers $\mathbb{N}$ under addition, and the monoid of integers $\mathbb{Z}$ under addition.[‡] First let's consider the function $f\colon \mathbb{N} \longrightarrow \mathbb{Z}$ with $f(n) = n$, so every natural number is mapped to itself as an integer. This "respects" addition in the following sense: we can add two numbers together before mapping them across, or we can add them together after mapping them across, and we get the same answer.

[†] I know, it's tricky when mathematicians say "recall" and you feel like it's something you've never heard of before. But what I mean is "I said this earlier and I'm hoping you have some recollection of it but if you don't it's in an earlier chapter for you to refer to".

[‡] We're taking the natural numbers to be the non-negative whole numbers; the integers are all of these and also the negative numbers.

$$\begin{array}{lrcl} \text{Adding them together before mapping gives this:} & f(n+m) & = & n+m \\ \text{Mapping them first before adding them gives this:} & f(n)+f(m) & = & n+m \end{array}$$

So it doesn't matter which way we do it, and this is one of the key conditions for being a "sensible" function between monoids. The condition is expressed abstractly like this: $\boxed{f(n+m) = f(n) + f(m).}$

Things To Think About

T 13.1 Another possible function $f\colon \mathbb{N} \longrightarrow \mathbb{Z}$ is given by $f(n) = 2n$. See if you can check that this also satisfies the above condition.

For this function $f(n+m) = 2(n+m)$, and $f(n) + f(m) = 2n + 2m$. These are equal, by the distributivity of multiplication over addition.

There are functions that are sensible and useful in other senses, just not in the sense of respecting addition. For example $f(n) = n + 1$ defines a perfectly sensible function $\mathbb{N} \longrightarrow \mathbb{Z}$, it just doesn't respect addition.[†] However, the function is sensible in the sense of being order-preserving (see Section 13.3).

Next we'll look at a function $\mathbb{N} \longrightarrow \mathbb{Z}$ that is useful and illuminating as it shows that we can "pair up" the natural numbers with the integers perfectly, even though on the face of it the natural numbers seem to be only half of the integers (the non-negative ones).

This function pairs the natural numbers with the integers by alternating between positive and negative numbers: the odd numbers are paired with positive numbers and the even numbers are paired with negative numbers (other than 0, which is paired with itself).[‡]

$$\begin{array}{ccr} 0 & \dashrightarrow & 0 \\ 1 & \dashrightarrow & 1 \\ 2 & \dashrightarrow & -1 \\ 3 & \dashrightarrow & 2 \\ 4 & \dashrightarrow & -2 \\ 5 & \dashrightarrow & 3 \\ 6 & \dashrightarrow & -3 \\ & \vdots & \end{array}$$

Things To Think About

T 13.2 See if you can show that this function does not respect addition.

The idea is that this is an unnatural process, switching between positives and negatives like that, so you might already feel like this is not respectful towards the structure of numbers. Of course, "respect" here doesn't mean politeness, it means something rigorous, but these visceral feelings can often be a decent starting point for the math. In fact, developing a visceral sense for things in a mathematical world is an important part of learning math. It's very hard to get anywhere without that, a bit like driving round a city relying *entirely* on GPS.

† $f(n+m) = n + m + 1$ but $f(n) + f(m) = n + 1 + m + 1 = n + m + 2$

‡ This is not a rigorous definition of the function but I hope the idea is clear. If you're interested in formality it's good practice to try expressing this function as a formula but I don't think the formula is particularly illuminating.

Anyway, we only have to find one place where this goes wrong in order to show that something has gone wrong, so let's take $n = 1$ and $m = 2$. Then $f(n + m) = f(3) = 2$ but $f(n) + f(m) = 1 + (-1) = 0$. So this function does *not* respect addition. However, it is still illuminating in the sense that it shows us that the natural numbers can be perfectly paired up with the integers. This is called a *bijection* and we'll come back to it in Chapter 14 as it gives us a pertinent notion of "sameness" for sets.

The above two non-examples were to illustrate that different functions can count as "sensible" or "illuminating" depending on which aspects of structure we are trying to illuminate. The idea of category theory is that while we are thinking about monoid structure, the "sensible" notion of morphism is the one that respects monoid structure.

There is one more aspect to a monoid structure other than addition (or more generally, the binary operation) — the identity. So if we're going to respect the entire monoid structure we need to respect that as well, which means that the identity should map to the identity. In our examples above, that means we need $f(0) = 0$, which is indeed the case.

Things To Think About

T 13.3 Mapping into the integers is a bit of a special case because it turns out that respecting addition forces a function to respect the identity. Can you prove that? We'll come back to it when we're talking about groups.

We have now basically seen the entire definition of a *monoid homomorphism* or just *monoid morphism*. "Morphism" is another word for "arrow" in a category, and "homomorphism" is a generic mathematical word for "morphism that leaves things the same", that is, a structure-respecting morphism, also sometimes referred to as structure-preserving.

Definition 13.1 Let A and B be monoids, and write the identity as 1 and the binary operation as $\circ$ in each case.[†] Then a *monoid homomorphism* from A to B is a function $f : A \longrightarrow B$ on the underlying sets, such that

- f respects $\circ$: for all $x, y \in A$ we have $f(x \circ y) = f(x) \circ f(y)$, and
- f respects the identity: $f(1) = 1$.

In general not every function on the underlying sets will respect the monoid structure, so this is a restriction to some special functions.

[†] Some authors would studiously always call the monoid $(A, 1, \circ)$ but I find that a little pedantic although it is arguably more correct. I think as long as it's unambiguous, we can just call the monoid A with impunity. Sometimes we say "by abuse of notation", like a confession.

Note on "obvious axioms"

Category theorists sometimes have a habit of referring to "the obvious axioms" rather than actually listing them, which can frustrate other people. However, if you get really into the spirit of category theory you'll start understanding what counts as "obvious axioms". When it comes to maps between sets-with-structure, the axioms ensuring that structure is preserved might be thought of as the "obvious axioms" for a map.

Our aim here was to form a category for the totality of monoids and their morphisms, and there are now a couple of things to check.

Things To Think About

T 13.4 What do we need to check to show that monoids and their homomorphisms form a category? Can you check it?

We need to check that there are identities and composition, and that they are unital and associative. Now, monoid morphisms are special functions between monoids. So if we can show that the usual identity functions and composition work, then the axioms will automatically hold, as they hold for all functions.

For the identities, we need to check that the identity function on any monoid is a monoid morphism. The identity function leaves everything the same, so it is indeed structure-preserving; that part is usually pretty easy when checking a category with structure-preserving maps.

Intuitively, composition should be immediate too: we start with composable functions f and g, and have to show that if they're each structure-preserving, then the composite is structure-preserving. It would be quite odd to do one structure-preserving thing followed by another and find that structure had not been preserved overall. But we can (and should) still check. The most confusing thing in what follows is probably notation, as there is the potential for using $\circ$ too many times. So I'm going to use $\circ$ for composition of the functions and $*$ for the binary operation in the monoids. First we show $g \circ f$ respects $*$:

$$\begin{aligned}
(g \circ f)(x * y) &= g\big(f(x * y)\big) && \text{by definition of } \circ \\
&= g\big(f(x) * f(y)\big) && \text{as } f \text{ respects } * \\
&= g\big(f(x)\big) * g\big(f(y)\big) && \text{as } g \text{ respects } * \\
&= (g \circ f)(x) * (g \circ f)(y) && \text{by definition of } \circ
\end{aligned}$$

Next we show that $g \circ f$ respects 1:

$$\begin{aligned}
(g \circ f)(1) &= g\big(f(1)\big) && \text{by definition of } \circ \\
&= g(1) && \text{as } f \text{ respects } 1 \\
&= 1 && \text{as } g \text{ respects } 1
\end{aligned}$$

So $g \circ f$ is a monoid morphism as required. Thus we have a category **Mnd** whose objects are all the monoids, and whose morphisms are all the monoid morphisms between them.

We quite often name categories like this: naming them after their objects, in abbreviation, in boldface. This is when it is sort of "obvious" what morphisms we are using. When it's not, then we might name the category after their morphisms; we'll see some examples of that later.

13.2 Groups

We have looked at homomorphisms of monoids as "structure-preserving functions", and we have looked at groups as monoids with an extra condition. So let's now try and combine those ideas to think about homomorphisms of groups. The idea is that we have been gradually building up structure on sets.

$$\begin{tikzcd}[row sep=large]
\text{sets} \arrow[d] & \text{no structure} \\
\text{monoids} \arrow[d] & \text{binary operation, identity} \\
\text{groups} & \text{binary operation, identity, inverses}
\end{tikzcd}$$

Note that at the moment those arrows I've drawn are just schematic arrows but, as you might have guessed, later we are going to see that they have a formal meaning and they are indeed arrows in some very large category, and that this process of building up structure is an intuitive idea that we make very precise.

Things To Think About

T 13.5 Can you invent the definition of "group homomorphism" for yourself, as a function respecting the group structure?

When we're dealing with groups it turns out to be redundant to ask for the identity to be preserved. It's also redundant to ask for the inverses to be preserved. Can you prove this?

A group is a monoid with some extra structure, so a group homomorphism should *at least* be a monoid homomorphism, but it should also preserve the extra structure. So we might think that a group homomorphism $f\colon A \longrightarrow B$ should be a function on the underlying sets such that

- the binary operation $\circ$ is respected: $\forall x, y \in a$, $f(x \circ y) = f(x) \circ f(y)$,
- the identity 1 is respected: $f(1) = 1$, and
- inverses are respected: $\forall x \in A$, $f(x^{-1}) = \big(f(x)\big)^{-1}$.

However it turns out that the last two conditions are redundant, that is, we can deduce them from the first one.

For identities, first note that $1 = 1 \circ 1$, so we have

$$\begin{array}{lrcll} & f(1) & = & f(1 \circ 1) & \text{applying } f \text{ to both sides} \\ & & = & f(1) \circ f(1) & \text{by the first condition} \\ \text{thus} & 1 & = & f(1) & \text{multiplying both sides by } (f(1))^{-1} \end{array}$$

Remember that multiplying anything by its inverse takes us back to the identity. I have sort of jumped over a step involving identities here. Also I've referred to $\circ$ as "multiplication" generically.

For the inverses, given any $x \in A$ we know $x \circ x^{-1} = 1$. So we have the following chain of logic. (I admit that this is one of those calculations that might not seem very illuminating overall, but I hope you can at least follow each step. It's more like a jigsaw puzzle where you put together the pieces in the only way that they fit, and then you're done.)

$$\begin{array}{lrcll} \text{First} & f(x \circ x^{-1}) & = & f(1) & \text{applying } f \text{ to both sides} \\ & & = & 1 & \text{as } f \text{ preserves identities} \\ \text{But also} & f(x \circ x^{-1}) & = & f(x) \circ f(x^{-1}) & \text{as } f \text{ respects } \circ \\ \text{thus} & f(x) \circ f(x^{-1}) & = & 1 & \text{combining the two parts} \\ \text{so} & f(x^{-1}) & = & (f(x))^{-1} & \end{array}$$

Note that the last step is by left-multiplying both sides of the equation by $(f(x))^{-1}$. (I specified *left*-multiplying as $\circ$ might not be commutative.)

Technicality on preserving identities

Proving that the condition on identities was redundant depended on the existence of inverses, which is why it doesn't happen for general monoids. When we mapped into the integers we were regarding it as a monoid but it secretly did have inverses, which is why identities were "automatically" preserved. There do exist monoid maps that respect the binary operation without respecting the identity. Here's a tiny example, in case you're curious.

- Let A be a monoid with two elements 1 and a, with $a^2 = 1$.
- Let B be a monoid with two elements 1 and b, with $b^2 = b$.

Now define a function $f\colon A \longrightarrow B$ by $f(1) = f(a) = b$. This does not preserve the identity, but it does respect the binary operation.

These monoids may seem a little contrived, but the first one involves an element that acts like a reflection, because if you do it twice you get back to the identity. The second one involves an element that acts like a projection,† because doing it twice is the same as doing it once. This example is not exactly important, it's just here to show that the condition on preserving identities really is necessary for monoids, and thus also for categories.

Things To Think About

T 13.6 Do we have to check anything else to show that groups and group homomorphisms form a category?

There isn't *really* anything else to check because this is really just the same as the definition of monoid morphism, and identities and composition will work just the same way. So we can declare we have a category **Grp** of groups and group homomorphisms.

13.3 Posets

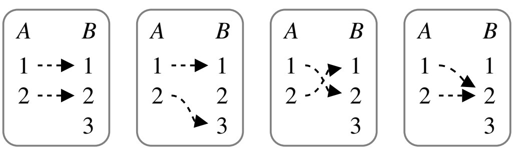
Posets (partially ordered sets) are another example we've seen of "sets with structure". In this case the structure was an ordering. We'll think about these in two ways: directly, and through the concept of structure-preserving maps.

Things To Think About

T 13.7 Consider the following totally ordered sets, where the ordering is $\leq$.

$$A = \{1, 2\}, \qquad B = \{1, 2, 3\}$$

The diagrams below depict some functions from A to B, showing the action on elements. Which ones do you think deserve to be called "order-preserving", and why? What visual features do you notice? What other functions can you think of that deserve to be called order-preserving? Can you draw all the possible functions (first working out how many there are) and decide which ones are order-preserving? What about for other examples of A and B?

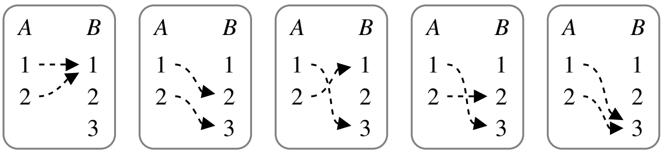


Can you make up a definition of "order-preserving function" for posets in general just by analogy with the structure-preserving maps we've seen already?

† A projection is like a shadow. If you're on a cartesian plane in XY-coordinates, you could project onto the X-axis by just taking the X-coordinate. If you then do it again you just get the same thing as you're already on the X-axis.

One way of thinking of "order-preserving" for a totally ordered set is to think about whether the elements stay in the same order after they've been mapped over to B. You could imagine two people in a queue in positions 1 and 2 in queue A, and then they're asked to move into queue B and the function shows the new positions they end up in. Are they in the same order or not? You can see this in the pictures from whether the dotted arrows cross over or not. In the first and second examples above they don't, and the two people do stay in the same order in the notional queue. In the third example they cross over, so the order has changed. The last example is a bit ambiguous — they've ended up in the same place. They haven't changed order but they haven't exactly stayed in order either.

We know from what we saw before that with 2 inputs and 3 outputs there should be 9 possible functions. Here are the other five:



The second one definitely keeps everything in the same order, the third and fourth definitely don't, and the first and last are ambiguous.

Sometimes when there's an ambiguous case, looking at the formalism can help us sort out a good way of thinking. Here we have posets with an ordering $\leq$ which is the "structure". A structure-preserving function $A \xrightarrow{f} B$ might then be reasonably thought of as one where for all $x, y \in A$ we have

$$x \leq y \implies f(x) \leq f(y).$$

Things To Think About

T 13.8 Does this match the definition we came up with from the pictures? Does it allow the ambiguous cases to count or not?

This formal definition includes the ambiguous cases as order-preserving, because the ordering includes equality as a possibility. If we didn't want to allow it we'd have to do something "stricter" and perhaps use $<$ instead. The trouble is that it's not such a categorical notion. (Recall that when we tried to express $<$ as arrows in a category we found we would not have any identities.)

Another way we could eliminate the ambiguous cases would be to put in a condition saying that different elements of A are not allowed to land on the same place in B, which is what we'd say if we really were asking people to move queue. In fact this is a sensible, interesting and illuminating condition to think about for any functions, not just order-preserving ones. It's called "injectivity" and we'll come back to it in Chapter 15.

Things To Think About

T 13.9 Do we have to check anything to show that we have a category of posets and order-preserving functions?

As with monoids and groups, we need to check that the identity function is order-preserving and that the composite of two order-preserving functions is order-preserving. And likewise nothing much happens here.

Let's quickly write out the part about composites: suppose we have maps $A \xrightarrow{f} B \xrightarrow{g} C$ which are order-preserving, and consider $x, y \in A$. Then

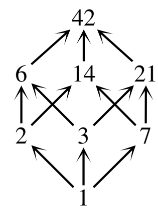
$$\begin{array}{rcll} x \leq y & \Longrightarrow & f(x) \leq f(y) & \text{since } f \text{ is order-preserving} \\ & \Longrightarrow & g(f(x)) \leq g(f(y)) & \text{since } g \text{ is order-preserving} \\ & \Longrightarrow & (g \circ f)(x) \leq (g \circ f)(y) & \text{by definition of composition} \end{array}$$

as required.

So we have a category **Poset** of posets and order-preserving functions.

Things To Think About

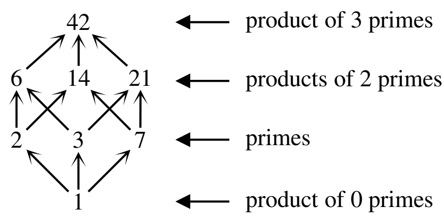
T 13.10 Think again about the poset of factors of 42 from Chapter 5. At that time we observed that $6 < 7$ but 6 is higher than 7 in this diagram. What two orderings are we considering on the same set of numbers here, and what map is it that is not order-preserving?



There are actually three orderings floating around in this diagram.

1. The one that we've actually drawn in arrows, where $a \longrightarrow b$ whenever a is a factor of b; remember this is written $a \mid b$.
2. The usual ordering according to size, $a \leq b$.
3. The physical levels on the page.

This last ordering really goes according to how many terms there are in the prime factorization of each number.



It's a bit long-winded to explain without some formal notation, so let's use this: given a whole number a, write $N(a)$ for the number of terms in its prime factorization.† Here is a table showing the values of N at each level of the cube. We will write $a \sqsupset b$ whenever $N(a) \leq N(b)$.

a	$N(a)$
42	3
6, 14, 21	2
2, 3, 7	1
1	0

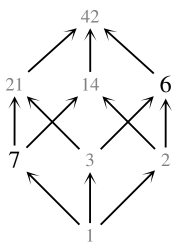
Now things are slightly tricky as we have one set but three different orderings on it. This is where it helps to declare what ordering we're using, so if we write A for the set of factors of 42 then we have the three posets shown on the right.

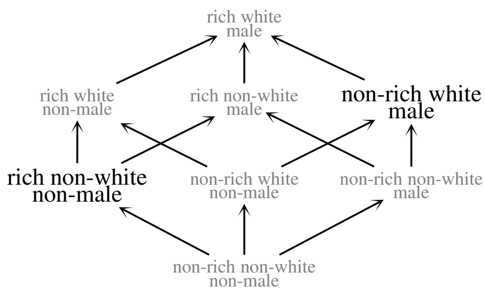
1. $(A, \mid)$
2. $(A, \leq)$
3. $(A, \sqsupset)$

We can now think about the identity function on A and ask whether or not it is order-preserving between these different posets. The issue of 6 being above 7 is formally this: $7 \sqsupset 6$ although $6 \leq 7$. Formally this is saying that the identity function on A is not an order-preserving map between these posets:

$$(A, \sqsupset) \longrightarrow (A, \leq).$$

In the case of the analogous cube of privilege it becomes a source of deep social conflict. Here are the analogous structures side by side. I have flipped the diagram of factors to emphasize which parts of the structure I'm seeing as analogous to 6 and 7.





The point here is that, just like with the factors of 42, there are two separate hierachies in the diagram of privilege: one by height on the page, which now represents the number of types of privilege someone has, and the other by *absolute* amounts of privilege or comfort that someone has in the world.

When ordered by number of types of privilege, poor white men have two types and so are "higher" on the page than rich black women; poor white men are in the position of 6 and rich black women are in the position of 7 with only one type of privilege. However, when it comes to absolute privilege I am

† For numbers with repeated prime factors we have to be careful to count the terms here because something like $2 \times 2 \times 3$ needs to count as having three terms. However here we can just think about the number of prime factors each number has, as they're all distinct.

sure that a sufficient amount of wealth can greatly mitigate for the very real disadvantages of being black and female, where no amount of maleness and whiteness can counteract being very poor, perhaps unemployed or homeless. Thus as with the factors of 42 we have an identity function that does not preserve order. This has helped me understand why some poor white men are particularly angry about the theory of privilege, as they are told they are very privileged, but they see people with technically less privilege doing better than them in society. So they think the theory of privilege is nonsense, because they don't feel the manifestation of any privilege.

I think it is much more productive to understand the source of this anger rather than simply get angry in return. This poset and the map that doesn't preserve order are how I found clarity around this issue, and I have found that the abstraction helps to keep emotional reactions out of the way and enable a less divisive discussion. I currently consider this to be the most important "application" of category theory I have found, even though it's really an application of categorical thinking rather than of a piece of category theory itself.

Incidentally the analogy between these structures is not just an informal similarity, it is a rigorous form of sameness between categories that we will come back to in Part Two of the book.

It is worth looking out for any non-order-preserving functions in life. Whenever you have two hierarchies on the same set of people and those hierarchies don't agree it is a potential cause of antagonism.

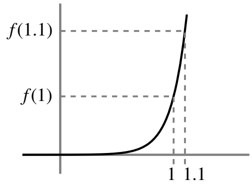
13.4 Topological spaces

We only briefly looked into topological spaces so we'll only briefly look into their structure-preserving maps here. Again, I think it is worth sketching some of the ideas here even though a rigorous treatment is beyond the scope of this book; see Appendix D for more details.

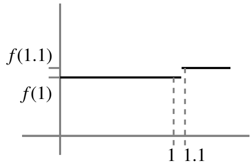
Topological spaces are sets with "a notion of closeness" (which is actually called *open sets*). So a sensible notion of function between them is going to be "functions that preserve closeness", which is the concept of a *continuous function*. The idea is that a function deserves to be called continuous if it doesn't break things apart. So for a function $A \longrightarrow B$ if two points were close together in A then they should not end up far apart in B. But note that "closeness" in topology isn't really to do with distance, it's more to do with connectedness.

For example let's think about functions $\mathbb{R} \longrightarrow \mathbb{R}$ so that they're a bit more familiar, and we can draw them as a graph and use visual intuition.

In this exponential graph $f(1)$ and $f(1.1)$ are much further apart than 0.1, because the graph of an exponential is so steep (and gets steeper all the time). However the points are connected so it sort of doesn't matter.



Whereas in this one $f(1)$ and $f(1.1)$ are closer together in terms of distance, but there's a gap in the graph in between meaning that in topological terms they're infinitely far apart as there's no way to get from one to the other.



Continuous functions (also called continuous maps) are typically defined in calculus via a tortured-looking formula that you might have seen at some point in your life, like this:

$$\forall a \in \mathbb{R}\ \ \forall \varepsilon > 0\ \ \exists \delta > 0\ \text{ s.t. }\ |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$$

I think this definition is brilliant, but unfortunately its symbols, the machinations involved in using it to prove anything, and the trauma associated with calculus classes can serve to obscure the elegance of its ideas and the point of continuous maps, which is the preservation of the closeness structure. That last point is all you really need to absorb for our purposes.

In calculus one goes to great lengths to prove that continuous functions compose to make continuous functions. That is very important, and together with the (easier) fact that the identity function is continuous, it gives us a category of topological spaces, written **Top**, with

- objects: all the topological spaces
- morphisms: a morphism $A \longrightarrow B$ is a continuous function $A \longrightarrow B$.

In practice topologists (and thus category theorists) often want to eliminate some really "pathological" weirdly behaving spaces from their study, and so we might restrict the category **Top** to include only better-behaved topological spaces, but keep all the morphisms between them.

In fact continuous functions are arguably not quite the point of topology. The definition of topological space is there to support the definition of continuous function, which is in turn there to support the definition of "continuous deformation". When we talked earlier about continuously nudging paths around to see if they were genuinely different or not, that was an example of a continuous deformation. We can do this thing to entire spaces to see whether they count as genuinely different or not. This is the idea behind the infamous topological result that "a coffee cup is the same as a doughnut". Note that this coffee cup has to have a handle and the doughnut has to have a hole. The idea is that if these were made of playdough you could turn one into the other by continuously squidging it around, without ever breaking or piercing it or sticking separate parts together. This is a more subtle notion of sameness appropriate for topology, and is technically called homotopy equivalence.

Homotopy is the notion of continuous deformation which is in some sense really the point of topology, but it requires more dimensions of thought than are available in the category of spaces and continuous maps. One way to deal with those higher dimensions is to go into higher-dimensional category theory, which we will briefly discuss at the end of the book. Another way is to try and work out how to detect traces of the higher dimensions in lower-dimensional algebra so that we can avoid creating new types of algebra to deal with it.

Algebraic topology largely takes the approach of detecting each dimension as a group structure, via *homotopy groups*, *homology* or *cohomology*.† This involves building some sort of relationship or relationships between the category **Top** of spaces and continuous maps, and the category **Grp** of groups and group homomorphisms. This means we need morphisms between categories.

13.5 Categories

We have been talking about large categories formed by totalities of mathematical structures. Our examples have included monoids, groups, posets and topological spaces. All of those are things we already had as small examples of mathematical structures.

That is, an individual monoid is an example of a category, as is any individual group, poset or topological space. We have a schematic diagram like this:

$$\begin{tikzcd}
\text{individual monoid} \arrow[r, "\text{special}", "\text{case of}"'] \arrow[d, "\text{totality}"'] & \text{individual category} \arrow[d, dashed, "\text{totality}"] \\
\text{category of monoids} \arrow[r, dashed] & ?
\end{tikzcd}$$

Things To Think About

T 13.11 What might you expect to find at the bottom right corner? What might the bottom dotted arrow represent?

This diagram is just schematic, that is, it's a diagram of ideas and thought processes, but it does hint to us that if the totality of monoids can be assembled into a category then we should be able to assemble a totality of categories themselves.‡ In that case monoid homomorphisms should be a special

† I've just included those terms in case you're interested in looking them up.

‡ We will use size restrictions to avoid Russell's paradox; see Appendix C for more details.

case of the arrows between categories. So should group homomorphisms and order-preserving functions. The totality of monoids should sit inside the totality of categories as something we might call a "sub-category".

We need a concept of "morphism between categories", and we can arrive at the definition by our usual method of thinking about structure-preserving maps. However, the situation is now more complicated because our underlying data isn't just a set: even if we restrict to small categories, we have a set of objects but also for every pair of objects a set of arrows from one to the other. As our data has two levels, our underlying function needs two levels as well: we need a function on objects and a function on arrows.

Things To Think About

T 13.12 Can you think of a sensible starting point for a definition of a morphism of categories $F\colon \mathbb{C} \longrightarrow \mathbb{D}$? It should map objects to objects and arrows to arrows, but if we start with an arrow $x \xrightarrow{f} y$ in $\mathbb{C}$ what should the source and target of $F(f)$ be in $\mathbb{D}$?

- We definitely want for every object $x \in \mathbb{C}$ an object $F(x) \in \mathbb{D}$.
- Given an arrow $x \xrightarrow{f} y$ in $\mathbb{C}$ we want an arrow $F(f)$ in $\mathbb{D}$ and it should live here
$$F(x) \xrightarrow{F(f)} F(y).$$

Sometimes we get a little bored of all these parentheses so we might omit them and just write Fx, Fy and Ff trusting that using uppercase and lowercase letters makes the point that F is applied to the lowercase letters.

We now need to think about what "structure-preserving" means.

Things To Think About

T 13.13 Can you come up with the definition of a morphism between categories, starting with the action of F on objects and arrows above, and then proceeding by making sure it is "structure-preserving"? To do this, you need to be clear what the structure of a category is: identities and composition. Once you've done this, can you check that this gives us a category of small categories and all morphisms between them?

These morphisms between categories are so important that they have an actual name rather than just "morphisms of categories": they are called *functors*. The property of preserving identities and composition is called *functoriality*. We will give the full definitions in Chapters 20 and 21. For now it's enough to keep in mind that functors are a good type of relationship for categories, via structure-preserving maps.

13.6 Matrices

Most of the examples we've seen so far have involved morphisms/arrows that are some sort of function or map, preserving structure. I want to stress that this need not be the case. We've seen a few small examples of that where the arrows were relations, or "assertions". In the case of factors an arrow $a \longrightarrow b$ was the *assertion* that a is a factor of b. In the case of ordered sets an arrow $a \longrightarrow b$ is the assertion $a \leq b$.

Another type of example was the category of natural numbers expressed as a monoid, where we had just one (dummy) object and the morphisms were the numbers themselves. One of my favorite examples is the category of matrices. In this category again it's not the objects that are really interesting but the morphisms: the morphisms are matrices. That is, we regard matrices as a sort of map.

A matrix is a grid of numbers like these examples. It doesn't have to be square — it could have a different number of rows and columns, like the second one.

$$\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \end{pmatrix}$$

The second one is called a 2×3 matrix as it has 2 rows and 3 columns; in general an $r \times c$ matrix has r rows and c columns. In basic matrices the entries are all numbers, but they could come from other suitable mathematical worlds; we won't go into that here.

One of the baffling things about matrices when you encounter them in school is how they get multiplied. Addition isn't too bad because you just add them entry by entry. But multiplication involves some slightly contorted-looking thing involving twisting the columns around onto the rows.

It is not my aim to explain matrix multiplication here, but if you do remember it you might remember that we could multiply the matrices shown, by sort of matching up the row and column shown, and continuing.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 7 & 1 & 0 \\ 0 & 2 & 4 & 2 \end{pmatrix}$$

The important point for current purposes is that the *width* of the first matrix has to match the *height* of the second. In general we can multiply an $a \times b$ matrix with a $b \times c$ matrix to produce an $a \times c$ matrix.

Things To Think About

T 13.14 Does this remind you of composition in a category? How?

In a bit of a leap of imagination we can regard an $a \times b$ matrix as a morphism $a \longrightarrow b$, in which case a $b \times c$ matrix is a morphism $b \longrightarrow c$ and the condition of composability matches the condition of when we can multiply matrices. We get a category **Mat** with

- objects: the natural numbers
- arrows: an arrow $a \longrightarrow b$ is an $a \times b$ matrix
- composition: matrix multiplication.

Things To Think About

T 13.15 If you've ever seen matrices before you might like to think about what the identity is and how we check the axioms.

This might seem like a weird category until we consider that one of the points of matrices is actually to encode linear maps between vector spaces.† In that way, matrices really do come from some kind of map, it's just that the encoding has taken them quite far away from looking like a map. That typically makes it very handy for computation (especially if you're telling a computer how to do it) but can be quite baffling for students if they're just told about matrices for no particular reason — like so much of math.

Incidentally, you might wonder if there's a category of vector spaces and linear maps, and whether there's a relationship (perhaps via a functor) between this category and the category of matrices. The answer is yes, and if you spontaneously wondered those things then that's fantastic: you're thinking like a category theorist.

Having a framework that works at all scales

This tour of math has gone on quite long enough now and I just want to end by remarking that we have seen examples of categories at many scales: small drawable ones, individual mathematical structures, and large totalities of mathematical structures. This way in which category theory can work at many different scales is one of its powerful aspects.

In the next part of the book we are going to investigate things that we actually do in categories and in category theory. In some sense it doesn't matter whether or not you understand or remember any of the examples we saw in this tour. If you're interested in abstract structures, the things we see in the next part can be of interest entirely in their own right.

† I mention all this in case you've heard of it; don't worry if you haven't.

PART TWO

DOING CATEGORY THEORY

In Part One we built up to the idea of categories, warmed up into mathematical formalism, and then met the definition of a category. We then took an Interlude to meet some examples of categories and see how various branches of mathematics could be seen in a categorical light. In Part Two we are going to "do" category theory. This means we are going to build on the basic definition of a category and think about particular types of structure we might be interested in, inside any given category. The point of the definition of a category is to give ourselves a framework for studying structure, so I would say that "doing category theory" means studying interesting structure, using the framework. The framework involves a certain amount of formality in order to achieve rigor, and so we are going to use that formality in what comes next. Part Two might therefore seem significantly harder than what came before, and I hope you will congratulate yourself for reading some quite advanced mathematics.